

1] What is NLP?

Natural Language Processing (NLP) is a branch of AI that enables computers to understand, interpret, & generate human language (text or speech).

It combines linguistics (language rules), machine learning & deep learning.

Ex: Chatbot, Google Translate, voice assistants etc

2] Types of NLP:

① Natural Language Understanding (NLU)

- Focus: Understanding meaning of text
- Tasks:
 - sentiment analysis
 - named entity recognition
 - Intent detection

② Natural Language Generation (NLG)

- Focus: Generating human-like text
- Tasks:
 - Text summarization
 - Chat responses
 - Content generation

3) Difference: NLP vs NLU vs NLG

Aspect	NLP	NLU	NLG
Definition	Overall field	Understanding text	Generating text
Goal	Process language	Extract meaning	Produce language
Example	Language translation	Detect sentiment	Write an article
Scope	Broad	Subset of NLP	Subset of NLP

NLP → Entire system

NLU → Input understanding

NLG → Output generation

4) NLP Life cycle:

① Data Collection: gather text data (tweets, reviews, etc)

② Text preprocessing: cleaning & preparing text

③ Feature extraction: convert text to numerical form

④ Model Building: Apply ML/DL models

⑤ Evaluation: Accuracy, precision, recall

⑥ Deployment: use in real-world applications

5] Techniques of NLP:

Common NLP techniques include:

- Tokenization
- Stemming
- Lemmatization
- Parts-of-Speech Tagging
- Named Entity Recognition (NER)
- Parsing
- Text Classification
- Sentiment Analysis

6] Challenges in NLP:

- ① Ambiguity: words with multiple meaning,
ex: "bank" (river bank / financial bank)
- ② Context understanding: same sentence different meaning depending on context
- ③ Sarcasm & Humor: Hard for machines to detect
- ④ Language diversity: multiple languages, dialects
- ⑤ Data quality: Noisy, incomplete or biased data
- ⑥ Grammar variations: informal text (social media slang)

7] Techniques available in NLP:

- ① Rule-Based Techniques: Use grammar rules
Ex: Regular expressions
- ② Statistical Techniques: Use probability models
Ex: Naive Bayes
- ③ Machine Learning techniques: Learn patterns from data
Ex: SVM, Decision Trees
- ④ Deep Learning techniques: Neural networks
Ex: RNN, LSTM, Transformers

8] Explanation of NLP techniques with examples

- ① Tokenization: splitting text into words or sentences
Ex: "I love AI"

Output: ["I", "love", "AI"]

- ② Stemming: Reducing words to root form (not always meaningful)

Ex: "running" → "run"

"studies" → "studi"

③ Lemmatization: Reducing words to meaningful base form

Ex: "better" → "good"

"running" → run

④ Part-of-speech (POS) Tagging: Assigning grammatical role

Ex: "She runs fast"

- She (Pronoun)
- runs (Verb)
- fast (Adverb)

⑤ Named Entity Recognition (NER):
Identifying entities like names, places

Ex: "Elon Musk founded Tesla"

- Elon Musk → person
- Tesla → organization

⑥ Sentiment Analysis:

Detecting emotion in text

Ex: "This movie is amazing" → Positive

"Worst experience ever" → Negative

⑦ Text classification:

Categorizing text

Ex: • Email → Spam / Not spam

• News → Sports / Politics